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TITLE: KINETICS AND SPECTROSCOPY OF KrF (B) AND Kr_2F ($4^2\Gamma$)

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Kinetics and Spectroscopy of KrF (B) and Kr_2F ($4^2\Gamma$)

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The absorption spectra corresponding to transitions from the $4^2\Gamma$ level to upper excited states of several rare gas-halide trimers (Kr_2F, Ar_2F and Xe_2Cl) have been measured at wavelengths ranging from 200 nm to as large as 800 nm. Absolute absorption cross sections for Kr_2F have been determined. Experiments are discussed that will measure the radiative lifetime of the KrF (B) state which is formed utilizing a sub-picosecond source. KrF B $\leftrightarrow$ C mixing, quenching rates and the B$\rightarrow$X stimulated emission cross section will be measured.

Introduction

A complete understanding of any molecular system requires knowledge of all reactions that produce or deplete the molecular species of interest in the system as well as the fundamental constants that describe those reactions. In the case of the rare gas-halide excimers (RGH), the species of interest include rare gas atoms and molecules, halogens, the excimers and the associated trimers. At pressures utilized in typical RGH laser mixtures, the trimers may be present in large quantities, and knowledge of their formation and decay mechanisms along with their spectroscopic behavior is essential in understanding the RGH system.

As an example, the RGH trimers are predicted to absorb in the ultraviolet in a manner similar to absorption by the ground state dimer ion.[1] In many cases, this absorption is predicted to be very strong at the RGH emission wavelength, so that the trimers could be strong absorbers of the desired excimer laser light. For the case of Kr_2F^* absorption at 248 nm (KrF B$\rightarrow$X emission wavelength), the absorption cross section was recently measured to be ~1×10^{-18} cm^2.[2-4] This value, though smaller than predicted,[5] is not insignificant, and Kr_2F^* absorption cannot be neglected in attempting to model the KrF system.

However, the RGH trimers are interesting in their own right, and lasing has been observed from Kr_2F and Xe_2Cl.[6,7] Its large emission bandwidth and potential tunability in the blue green portion of the spectrum have made Xe_2Cl a candidate in the blue green laser program. Measuring the absorption spectra of the RGH trimers becomes all the more important as they are considered for various scientific and military applications.

Up until the last few years, only limited absolute absorption cross section measurements on Xe_2Cl and Kr_2F were made, and only at specific wavelengths.[8,9] Recently, Geohegan and Eden[4] attempted to continuously measure the Kr_2F and Xe_2Cl absorption spectra using a tunable dye laser, and although low frequency doubling efficiencies kept their measurements above 340 nm, the resulting spectra decreased in absorption as the dye laser wavelength increased, and several interesting features were observed. However, a complete continuous absorption spectrum had not been measured for any of the trimers until this past year when the Kr_2F ($4^2\Gamma$) absorption spectrum from 255 nm to 455 nm was published.[10] The work reported here is an extension of that result to wavelengths as low as 200 nm and as large as 800 nm, and also includes results on Ar_2F and Xe_2Cl.

In addition to measuring the relative spectral dependence of the absorption, a refined technique for measuring absolute cross sections has given a more reliable value for the Kr_2F absorption cross section. In contrast to previous absolute cross section measurements, the cross sectional area of the probe beam was continuously monitored using a linear pyroelectric array, and the probe beam was telescoped down to a smaller size and collimated through the cell (unfocussed).

Cross Section Measurements

In measuring the spectral dependence of the trimer absorption, a probe beam was sampled before and after an absorber was created in a specified length L, and the ratio of intensities obeyed Beer's law,

$$\frac{I}{I_0} = e^{-N\sigma L} \qquad (1)$$

where I_0 is the incident or unperturbed intensity, I is the intensity detected after absorption takes place over the length L, σ is the absorption cross section and N is the density of absorbers. This assumes that the absorber density is not depleted by the probe beam. In order to determine σ, the absolute density of absorbers must be ascertained, and therefore, the absorption coefficient, $\alpha(\lambda)$, equal to $N\sigma(\lambda)$, was determined.

Absolute cross sections for Kr_2F were measured using the fluorescence suppression technique.[11] An excited state was created using laser excitation, which resulted in the production of only a few species. That state's absorption cross section was determined by examining its fluorescence in the presence and absence of a probe beam of a particular wavelength. Since absorption reduced the density of absorbers, the

fluorescence from the excited absorbers was suppressed. This is described mathematically in the following analysis for the case of Kr_2F^* absorption. The rate equation that describes the time rate of change of Kr_2F^* is given by the following:

$$\frac{d[Kr_2F^*]}{dt} = P[S] - [Kr_2F^*]\left\{\frac{1}{\tau} + k[Q] + \frac{\sigma I}{\hbar\omega}\right\} \qquad (2)$$

where P is the production rate from species S, τ is the Kr_2F^* radiative lifetime, k is the rate constant for quenching of Kr_2F^* by the quencher Q, σ is the wavelength dependent absorption cross section, I is the probe laser intensity in W/cm^2, $\hbar\omega$ is the laser photon energy and brackets denote particle densities. This assumes that one operates in a regime where there is no appreciable loss of laser photons and stimulated emission from Kr_2F^* does not take place. It is also assumed that absorption takes place to a level which immediately dissociates or relaxes quickly and that reformation of Kr_2F^* can be ignored. If the pumping rate is fast compared to the loss rate and if one waits to fire the probe laser until P is small, then the Kr_2F^* density can be solved for by neglecting P.

$$[Kr_2F^*(t)] = [Kr_2F^*(t_o)] \exp\left\{-\frac{\sigma}{\hbar\omega}\int_{t_o}^{t} I(t')\,dt'\right\} \cdot \exp\left\{-\frac{t - t_o}{\tau'}\right\} \qquad (3)$$

The effective lifetime of the excited state, τ', is defined in the following way:

$$\frac{1}{\tau'} = \frac{1}{\tau} + k[Q] \qquad (4)$$

and t_o is the time at which the probe laser is fired. If the laser pulse length is T_L, the ratio of the Kr_2F^* density at time $t_o + T_L$ with and without firing the probe laser, or the fractional fluorescence (FF), is

$$FF = \frac{[Kr_2F^*(t_o+T_L,\phi)]}{[Kr_2F^*(t_o+T_L,\phi = 0)]} = e^{-\sigma\phi} \qquad (5)$$

where the laser fluence, ϕ, in photons per cm^2, is equal to $E/\hbar\omega A$, and E, the total energy in the incident laser pulse, is given by

$$E = A\int_{t_o}^{t_o+T_L} I(t)\,dt \qquad (6)$$

The cross section, then, is determined by measuring the ratio of the fluorescence with and without the action of a probe beam as a function of laser fluence. In order to make an accurate determination of σ, one must be able to measure accurately the cross sectional area of the probe beam, the energy that is contained in the pulse and the amount of suppression that takes place. These are the major sources of uncertainty in this measurement.

Experimental Apparatus

A. TASS Set-up

The Transient Absorption Spectroscopy Set-up (TASS) is shown schematically in Fig. 1. Large densities of absorbers were created in a small discharge laser (EMG-50) having MgF_2 windows and a discharge volume of ~24 cm^3. Light from a pulsed, xenon flashlamp was collimated by a short focal length (FL) fused silica lens, apertured and passed through the discharge before being focused with an achromatic lens onto the 25-μm entrance slit of a 0.25-m spectrometer. The spectrometer dispersed the light onto a gated, intensified, ultraviolet-sensitive diode array, and a 300-groove/mm grating gave a 200-nm range over the active portion of the diode array. A 20-ns gate pulse was applied to the array detector at the peak of the 300-ns lamp pulse. Timing between the lamp, discharge device and high voltage pulser was controlled with a delay generator, and the detection of discharge fluorescence with a photodiode (PD) allowed the drift of the thyratron-switched discharge to be observed and corrected for.

The discharge device was typically filled with 400 to 700 mBar of rare gas (xenon, krypton or argon), 5 mBar of halogen (F_2 or HCl) and a balance of helium to a total pressure of 2.5 Bar. Since excessive rare gas pressures resulted in arcing, the device was run at the highest rare gas pressure that would give uniform, reliable discharges, so that three body formation rates of the trimers would be maximized while maintaining shot to shot reproducibility.

The detector controller was interfaced to a computer that stored and analyzed the data. Each individual absorption spectrum required three data runs of typically 600 shots per run at a repetition rate of 10 Hz. The first run (I_{Abs}) consisted of acquiring both the lamp light and the discharge fluorescence, and it contained the absorption information. The next two runs (I_{Lmp} and I_{Fl}) collected lamp light and discharge fluorescence separately. Subtraction of background

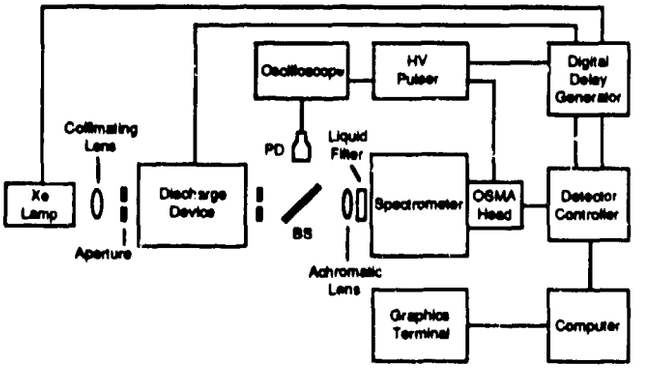


FIG. 1. Schematic of the Transient Absorption Spectroscopy Set-up (TASS). The heart of the system is an Optical Spectrometric Multichannel Analyzer (OSMA) which contains a gated head and a detector controller. The system is run by a computer which also stores and processes data.

light was automatic. Absorption was calculated in the following way:

$$\alpha(\lambda) = -\frac{1}{L} \ln\left\{\frac{I_{Abs}(\lambda) - I_{FI}(\lambda)}{I_{Lmp}(\lambda)}\right\} \quad (7)$$

where $\alpha(\lambda)$ was the net absorption in m^{-1} and L was the discharge length (0.2 m).

B. LEAKS Set-Up

Figure 2 is a schematic diagram of the experimental apparatus used to measure absolute cross sections (LEAKSS, or Laser Excitation for Absorption, Kinetics and Spectroscopy Set-up). Two excimer lasers (Lambda Physik EMG-201E and 101-MSC) were aligned so that their beams were counter-propagating. Both lasers were configured as oscillators with standard optics and produced 15- to 25-ns full width at half maximum beams which were unpolarized. The lasers were triggered at 10 Hz by a trigger generator, and each had a firing jitter of ±2 ns. The 201 (pump laser) ran on an argon fluoride gas mix (193 nm) and produced 300-mJ pulses with a shot to shot variation of 20%. The beam was focussed by a 7.5 cm FL cylindrical lens to a line at the center of a quartz cell. A gas handling system typically filled the cell with 1 Torr of electronic grade fluorine and 500 or 1000 Torr of research grade krypton. The energy of the ArF pulse was monitored with a quartz beamsplitter (BS) set to give a small (<5°) angle of incidence and it reflected ~8% of the beam onto an energy detector.

The probe laser was filled with either XeF (351 nm), N_2 (337 nm), XeCl (308 nm), KrF (248 nm), KrCl (222 nm) or ArF gas mixtures and fired an adjustable time delay, t_o, after the pump laser. After passing through a methanol or isopropyl alcohol liquid filter (except in the case of ArF mixes in both lasers),

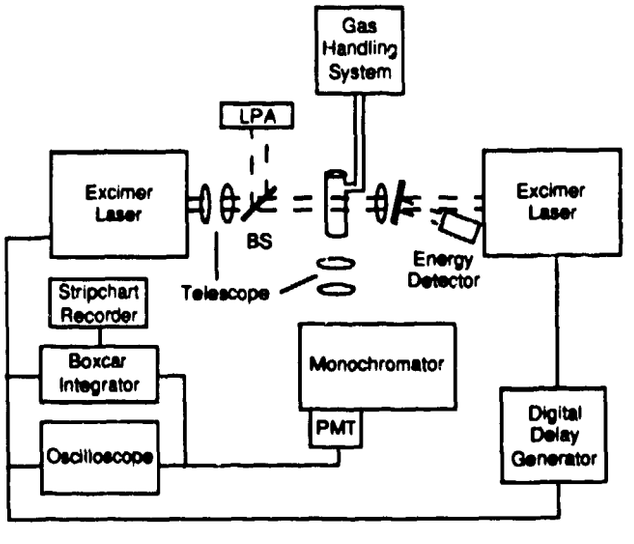


FIG. 2. Schematic diagram of the Laser Excitation for Absorption, Kinetics and Spectroscopy Set-up (LEAKSS). Absorbing species are created using the laser on the right, while the laser on the left produces the probe pulse. The liquid filter mentioned in the text (located at the output of the probe laser) is not shown.

the probe beam was directed through 10 cm and 2.5 cm FL cylindrical lenses which reduced the cross sectional area of the beam while retaining its collimation. The purpose of the liquid filter was to prevent the ArF beam from interfering with the gas chemistry inside the probe laser. This had been shown to have a deleterious effect on the probe laser's output.[12] A quartz BS reflected ~8% of the probe beam onto a Spiricon linear pyroelectric array (LPA) which was positioned the same distance from the BS as the quartz cell. In this way, the probe beam cross sectional area in the center of the cell could be determined exactly. The probe laser pulse energy was also monitored using another BS in combination with a dielectric full reflector and energy detector (not shown in the figure). The mirror helped to reduce the number of pump laser photons which reached the probe beam detector, although it did not altogether eliminate them. Output from the energy detector was displayed on both a readout unit and a stripchart recorder, allowing stray pump energy to be accurately subtracted out of the probe energy measurement.

Fluorescence from excited species in the cell was collimated and focussed onto the 200-μm entrance slit of a 27-cm monochromator and a portion of the dispersed fluorescence was detected by a Hamamatsu R943-02 photomultiplier (PMT). Kr_2F^* fluorescence was monitored at 400 nm (the peak in the emission band) and care was taken to ensure that no stray light from either laser was detected. The PMT signal was displayed on a Tektronix 7104 oscilloscope and sampled by a boxcar integrator. The rising edge of the 5-ns boxcar aperture was set to coincide with the end of the probe laser pulse (time $t_o + T_L$ in eqn. (5)). The boxcar's averaged output, which was proportional to the Kr_2F^* density at $t_o + T_L$, was recorded by the stripchart recorder.

Passing the probe beam through two cylindrical lenses with FLs of 10 and 2.5 cm resulted in a four fold reduction in the width of the beam. This provided fluence levels that were more than sufficient to observe 50% suppressions in fluorescence for output energies in the 50 to 100 mJ range. However, in the cases of N_2 and KrCl, the laser output was only ~10 mJ resulting in maximum suppressions of ~20%. The separation of the lenses was chosen to give a well collimated beam and had to be adjusted for each probe laser gas mix.

Data Taking and Results

A. Spectral Measurements

The dependence of the Kr_2F ($4^2\Gamma$) absorption on wavelength is seen in Fig. 3, as determined from the TASS set-up. The spectra were recorded at a delay of 60 ns between the start of the discharge and the end of the gate pulse. Three Kr_2F ($4^2\Gamma$) absorption features are evident in the figure. The ($4^2\Gamma \rightarrow 9^2\Gamma$) band peaks at 315 nm and has a half width of 85 nm. The ($4^2\Gamma \rightarrow 6^2\Gamma$) band peaks at ~690 nm, and a third band, which is definitely due to Kr_2F ($4^2\Gamma$) absorption, peaks at 222 nm and is much narrower than the ($4^2\Gamma \rightarrow 9^2\Gamma$) band. Also seen in the figure are the Kr_2^+ [$1(1/2)_u \rightarrow 2_1(1/2)_g$] and [$1(1/2)_u \rightarrow 1(1/2)_g$] absorption spectra that were predicted by Wadt.[13] These results demonstrate the near identical absorption features of the trimers

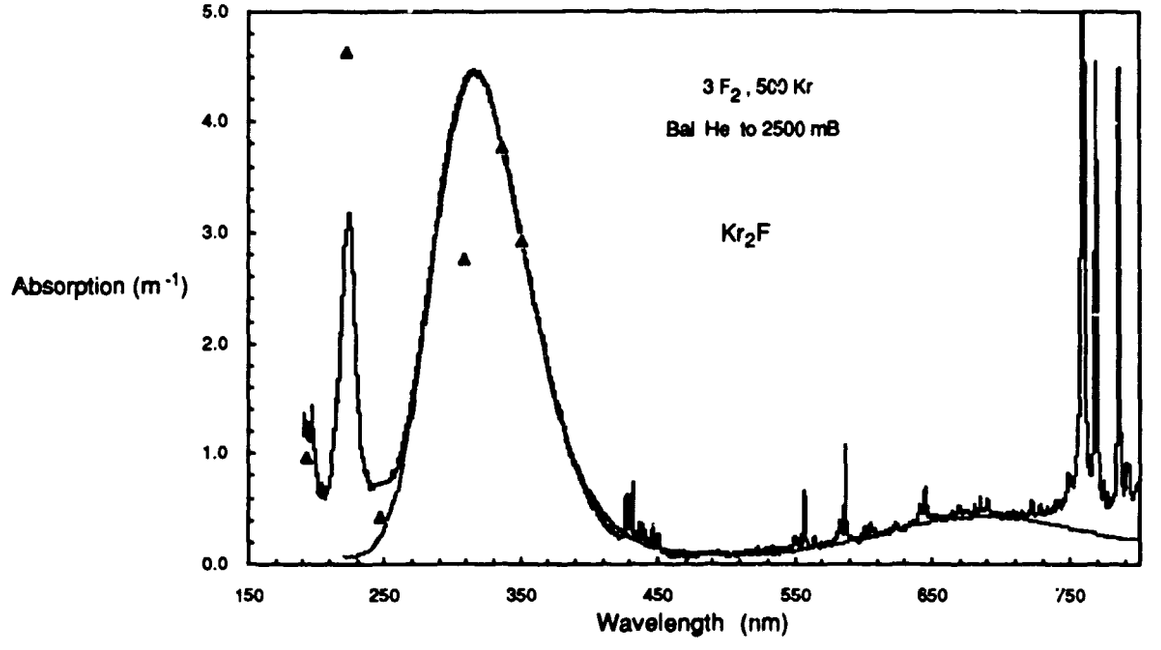


FIG. 3. Absorption spectrum of Kr_2F $4^2\Gamma$ and atomic absorbers as determined from TASS. Also displayed as triangles (▲) are the absolute cross section data determined from LEAKSS, and the dotted line is a combination of the Kr_2^+ $[1(1/2)_u \rightarrow 2(1/2)_g]$ and $[1(1/2)_u \rightarrow 1(1/2)_g]$ absorption bands predicted by Wadt. The theoretical data has been shifted slightly so that the peaks in the Kr_2F $(4^2\Gamma \rightarrow 9^2\Gamma)$ and Kr_2^+ $[1(1/2)_u \rightarrow 2(1/2)_g]$ absorption bands overlap. The absolute cross section data has also been normalized. The gas mixture was 3 mBar F_2, 500 mBar Kr and 1997 mBar He. The $4^2\Gamma \rightarrow 9^2\Gamma$ absorption band peaks at 315 nm and a second, narrower band is present to the blue, peaking at 224 nm. The narrow lines seen above 425 nm are metastable absorption features.

compared to the rare gas dimer ions, except for the third band seen at shorter wavelengths. The narrow lines seen between 425 nm and 800 nm are atomic absorption features.

Figures 4 and 5 give the absorption spectra of Ar_2F and Xe_2Cl. In both cases, the $(4^2\Gamma \rightarrow 9^2\Gamma)$ band is observed as well as the narrower band to the blue. Also shown in Fig. 5 are absorption data taken by McCown *et al.*[9] using the fluorescence suppression technique. The size of the cross section at 193 nm suggests that the band peaking at ~200 nm is real, and that a progression of bands may exist in the ultraviolet.

B. Absolute Cross Sections

The argon fluoride laser[14] was used as a means to excite krypton atoms from their ground state to the excited $6p(3/2)_2$ level in a resonant, two photon process. Quenching takes place rapidly to relax the excited atoms to the metastable level, and the formation of the KrF (B) state by the harpoon reaction is also fast. Kr_2F $(4^2\Gamma)$ is formed in a three body process:

$$KrF(B) + 2\,Kr \rightarrow Kr_2F(4^2\Gamma) + Kr \qquad (8)$$

Following excitation by the ArF pulse, the KrF (B→X) fluorescence peaked ~32 ns after initiation while the Kr_2F $(4^2\Gamma \rightarrow 1^2\Gamma)$ fluorescence peaked ~30 ns later. In fluorescence suppression measurements, the probe laser was triggered 70 ns after the ArF laser, and the 5-ns boxcar aperture was opened at the end of the probe laser pulse, typically 25 to 30 ns after the probe laser was fired. Approximately 500 shots were taken in the absence of the probe pulse, and the average of the fluorescence intensity was recorded by the boxcar-strip chart combination. This was repeated while firing the probe laser, and the fractional fluorescence was determined by dividing the intensities. The experiment was repeated at a variety of fluence levels, providing a set of ordered pairs of fractional fluorescence versus fluence. The fluence was varied by placing quartz flats (suprasil or optosil) at the output aperture of the probe laser, or by varying the laser gas mix or charging voltage. In this way, the fluence could be reduced by over an order of magnitude. The uncertainty in each data point arises from uncertainties of less than 10% in the energy measurement, 5% in the beam area measurement and 3% in the fractional fluorescence calculation.

A semilog plot of FF versus photon fluence is shown in Fig. 6. The probe laser wavelength was 308 nm, and the straight line drawn is a linear least squares fit to the data. The absolute value of the slope of the line, which is equal to the absorption cross section, is $(5.1 \pm 0.5) \times 10^{-18}$ cm^2. Data was taken at each of the other probe laser output wavelengths, and the absorption cross sections were determined. Results are presented in Table 1 and shown graphically in Fig. 3, along with the data obtained from TASS. The overall uncertainty in the cross section measurement is the standard deviation from the

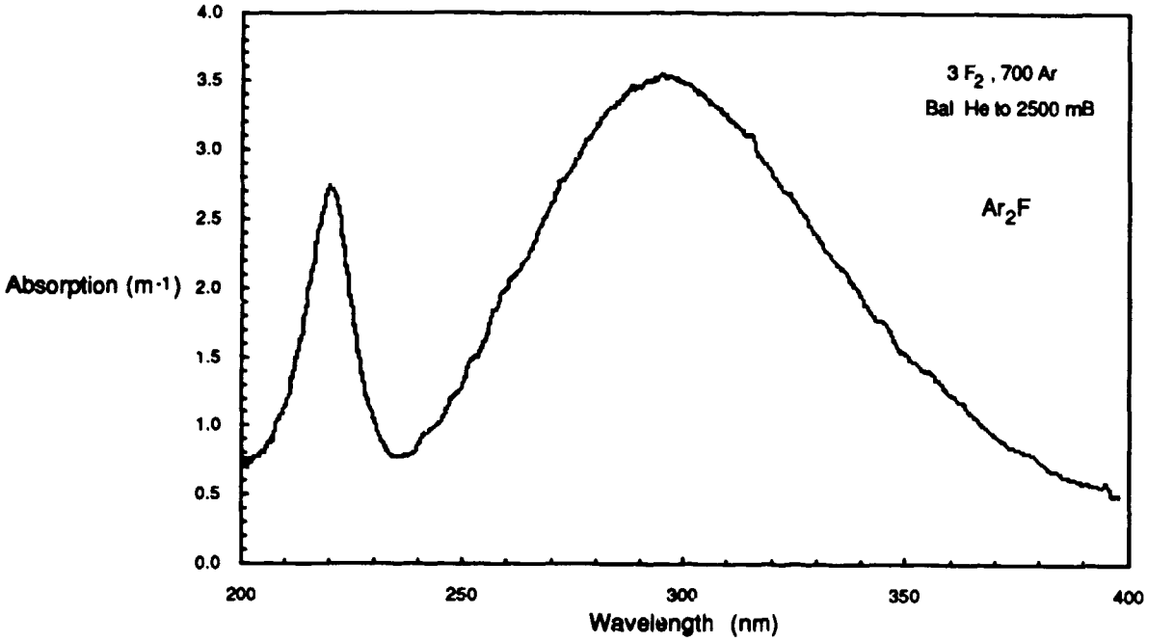


FIG. 4. Absorption spectrum of Ar_2F ($4^2\Gamma$). The gas mixture was 3 mBar F_2, 700 mBar Ar and 1797 mBar He. The ($4^2\Gamma \rightarrow 9^2\Gamma$) absorption band peaks at 295 nm and a second, narrower band is present to the blue, peaking at 220 nm.

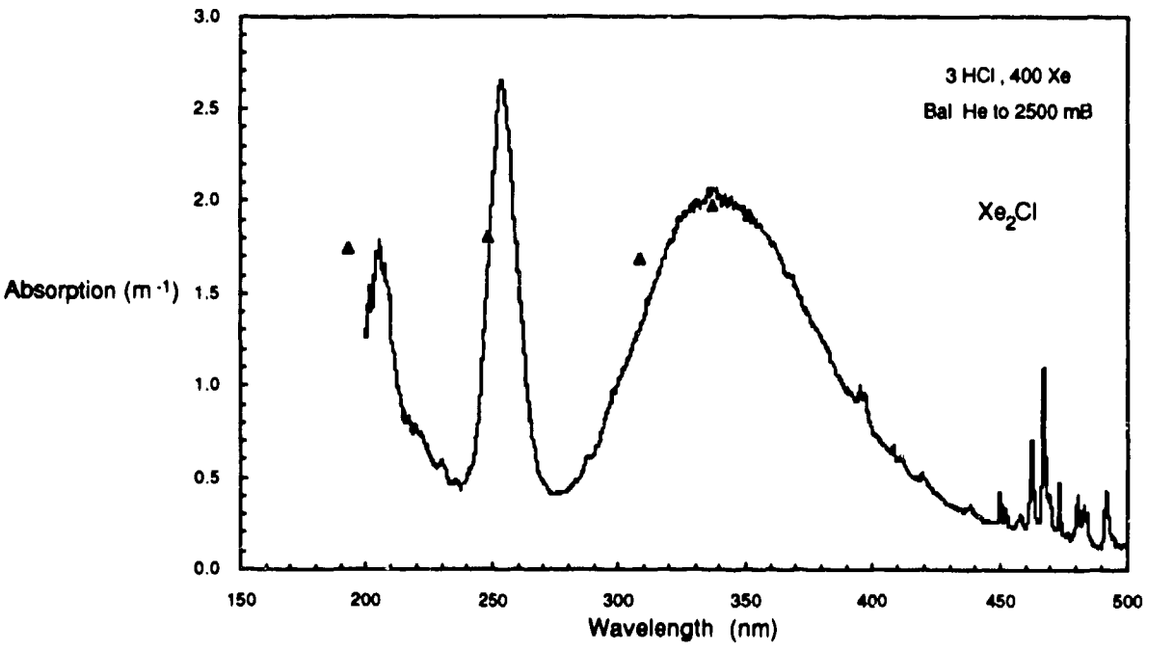


FIG. 5. Absorption spectrum of Xe_2Cl ($4^2\Gamma$) and atomic absorbers. Also displayed as triangles (▲) are the data taken by McCown *et al.* (Ref. 9) which has been normalized. The gas mixture was 3 mBar HCl, 400 mBar Xe and 2097 mBar He. The ($4^2\Gamma \rightarrow 9^2\Gamma$) absorption band peaks at 337 nm and a second, narrower band is present to the blue, peaking at 254 nm. A third band is also present peaking at $\lambda \leq 200$ nm, implying the existence of a progression of bands. The narrow lines seen above 450 nm are metastable absorption features.

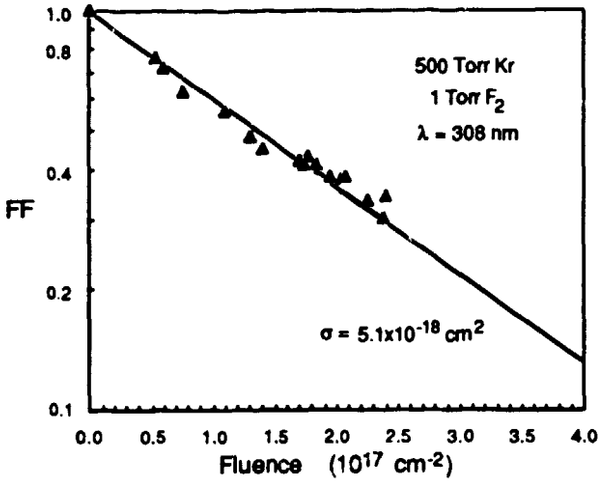


FIG. 6. Semi-log plot of the fractional fluorescence (FF; defined in text) as a function of the XeCl photon fluence. The triangles (▲) are data points and the straight line is a linear least squares fit to the data, giving a cross section of 5.1x10^{-18} cm^2.

mean, as determined from the least squares fit. The uncertainty is much larger in the N_2 and KrCl cases, since low fluences resulted in more limited data.

Discussion

Powell and Jancaitis[15] performed absorption experiments on electron-beam pumped Ar/Kr/F_2 mixtures using a KrF laser whose output was raman-shifted in H_2 and D_2, simultaneously producing 12 lines ranging from 206 nm to 423 nm. They determined from computer model predictions that Kr_2F^* was the dominant species present at the time the probe laser was fired, and attributed their results to Kr_2F^* absorption. Their spectrum is similar to the one shown in Fig. 3, and they also observed an increase in the cross section at 225 nm, but were unable to make a definite identification of the absorber at that wavelength. It is clear that this is also due to Kr_2F ($4^2\Gamma$) absorption.

From these results, it is unclear whether Kr_2F^* absorption at 248 nm excites a bound upper level, a state that immediately dissociates, or high vibrational levels of an excited state that quickly relax. Recent studies of Kr_2F^* absorption have shown that absorption at wavelengths greater than 280 nm results in photodissociation of Kr_2F^* and the immediate formation of the KrF (B) state,[16] while absorption at wavelengths below 250 nm do not appear to do so. The measurement at 248 nm is complicated by the effect of stimulated emission taking place which reduces the KrF (B) population.

TABLE I. Kr_2F ($4^2\Gamma$) absorption cross sections.

λ (nm)	σ (10^{-18} cm^2)
193	1.8
222	8.6
248	0.8
308	5.1
337	7.0
351	5.4

The absorption band from 250 to 500 nm is seen to be bound → free, supporting the contention that photodissociation takes place, leaving an excited KrF molecule and a krypton atom. The band centered at 224 nm, though much narrower, also appears to be bound → free, and the absorption products have not as yet been identified. Although the limit of the OSMA detection system is 200 nm, a new vacuum ultraviolet detector head will extend measurements below 200 nm, allowing a comparison to the 193 nm data point to be made.

The ($4^2\Gamma \rightarrow 9^2\Gamma$) absorption band is narrower in the discretely measured spectrum using the fluorescence suppression technique. The cause of this is not currently known, and several measurements were made to verify the accuracy of the 308 nm cross section, including performing the experiment at krypton pressures of 500 and 1000 Torr. Since photodissociation of Kr_2F^* takes place at this wavelength resulting in the immediate formation of KrF (B), doubling the krypton pressure increases the formation rate of Kr_2F^* from the newly formed KrF by a factor of four. If this increase in Kr_2F^* density were to have interfered with the fluorescence suppression measurement, it would have resulted in the measurement of a smaller cross section at 1000 Torr. However, the results at the two pressures were identical. The possibility of there being an error in the value of the absorption cross section due to neglecting the feeding term in eqn. (2) (Kr_2F reformation from the KrF photodissociation product) is currently under investigation.[17]

In addition to examining pressure effects, the effect of pump laser wavelength on the cross section measurement has been examined by Geohegan.[18] Since the ArF laser photons excite krypton atoms, there is another pathway to form Kr_2F^*; either through the KrF excimer, or by the process

$$Kr^* + 2\,Kr \rightarrow Kr_2^* \qquad (9)$$

$$Kr_2^* + F_2 \rightarrow Kr_2F^* + F \qquad (10)$$

Since under the general conditions of low Kr density the formation rate of KrF from the krypton metastable is much larger than that of reaction (9), it has been assumed that the Kr_2F^* is predominantly formed through KrF. However, the presence of large amounts of Kr_2^* in the cell would complicate the cross section determination, because it may not be true that the pumping term in eqn. (2) is small when the probe laser is fired. In Geohegan's measurement, he found that the 351 nm absorption cross section measured under conditions of KrF pumping, which results in photodissociation of F_2 and photoassociation of KrF directly, was within the experimental uncertainty of the value obtained in the ArF pumped case. Therefore, it is extremely unlikely that error is introduced by pumping with ArF.

Kinetics Experiments

Because of their short radiative lifetimes, it is difficult to perform excimer kinetics experiments using nanosecond pulse length lasers. The development of picosecond excimer lasers[19] has provided a means of creating or probing excited states 'instantaneously', allowing kinetics measurements to be made without having to deconvolve the laser pulse width. This is especially important in measuring the (B) state radiative lifetime, which in the case of KrF [20] and ArF [21] is less than 10 ns.

Experiments are now in progress using a sub-picosecond KrF laser system to photoassociate the KrF (B) state from Kr-F pairs. Free fluorine atoms are created several hundred microseconds before the arrival of the sub-picosecond pulse by dissociating molecular fluorine with either a KrF or an XeCl laser pulse. Thus, KrF (B) molecules are formed in sub-picosecond time frames allowing the radiative lifetime and quenching to be determined by observing fluorescence decay rates as a function of F_2 density. Mixing between (B) and (C) is examined by measuring the fluorescence decay rates of both excimer levels as a function of Kr and Ar densities. By raman-shifting the 248-nm pulses in hydrogen to produce 277-nm pulses, it is expected that KrF (C) will also be photoassociated in a low pressure environment, and the radiative lifetime and quenching rate constants will be measured. The fluorescence suppression technique can be used to determine the KrF (B) stimulated emission cross section.

Conclusions

Absolute absorption cross sections have been determined which correspond to upward transitions from the Kr_2F ($4^2\Gamma$) state. A technique which determines exact fluences using a pyroelectric array has been developed which eliminates the major uncertainly in past measurements. The absolute cross section measurements may have been influenced by re-feeding of the Kr_2F^*, and this is currently being investigated. Although the 250-nm to 500-nm absorption band determined from optically pumping Kr_2F is spectrally narrower than that obtained from TASS, the data is reproducible and suggests that there may be a difference between optical pumping and discharge or electron beam pumping which manifests itself in the absorption measurement.

Continuous measurements of the absorption cross section for the trimers Kr_2F, Ar_2F and Xe_2Cl have been made using a transient absorption set-up. The ($4^2\Gamma \rightarrow 9^2\Gamma$) band has been observed for each of the trimers, as well as another band to the blue that has not been predicted to exist. Experiments are currently in progress that will extend these measurements down to as low as 125 nm.

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